Automation with Workflows in MISP

Short version

Sami Mokaddem

MISP Project
https://www.misp-project.org/



Content of the presentation

1. Automation in MISP
2. MISP Workflows
   - Fundamentals
   - Demo with examples
   - Using the system
   - How it can be extended

Content of the presentation

1. Automation in MISP
2. MISP Workflows
   - Fundamentals
   - Demo with examples
   - Using the system
   - How it can be extended

Automation in MISP: What already exists?

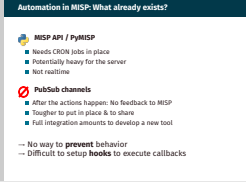
MISP API / PyMISP

- Needs CRON Jobs in place
- Potentially heavy for the server
- Not realtime

PubSub channels

- After the actions happen: No feedback to MISP
- Tougher to put in place & to share
- Full integration amounts to develop a new tool

→ No way to **prevent** behavior
→ Difficult to setup **hooks** to execute callbacks



Simple automation in MISP made easy



- **Visual** dataflow programming
- **Drag & Drop** editor
- Flexible **Plug & Play** system
- **Share** workflows, **debug** and **replay**

Simple automation in MISP made easy
MiSP
Threat Sharing
- **Visual** dataflow programming
- **Drag & Drop** editor
- Flexible **Plug & Play** system
- **Share** workflows, **debug** and **replay**

Example of use-cases

- **Notification** on specifc actions
  - New events matching criteria
  - New users
  - Automated alerts for high-priority IOCs
- **Extend** existing MISP behavior
  - Push data to another system
  - Automatic enrichment
  - Sanity check to block publishing / sharing
  - Curation pipelines
- **Hook** capabilities
  - Assign tasks and notify incident response team members
- ...

Example of use-cases
- **Notification** on specifc actions
  - New events matching criteria
  - New users
  - Automated alerts for high-priority IOCs
- **Extend** existing MISP behavior
  - Push data to another system
  - Automatic enrichment
  - Sanity check to block publishing / sharing
  - Curation pipelines
- **Hook** capabilities
  - Assign tasks and notify incident response team members
- ...

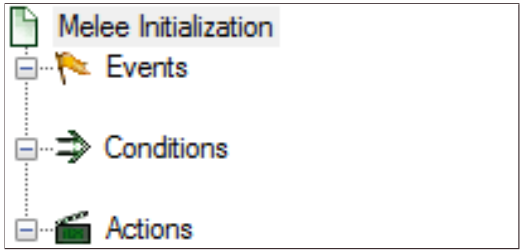
Workflow - Fundamentals

Objective: Start with the foundation to understand the basics



Workflow - Fundamentals
Objective: Start with the foundation to understand the basics

How does it work



1. An **event** happens in MISP
2. *(optional)* Check if all **conditions** are satisfied
3. Execute all **actions**
   - May prevent MISP to complete its original event

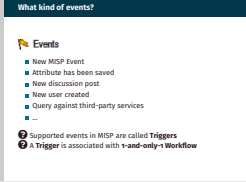
How does it work
Melee Initialization
Events
Conditions
Actions
1. An **event** happens in MISP
2. *(optional)* Check if all **conditions** are satisfied
3. Execute all **actions**
   - May prevent MISP to complete its original event

What kind of events?

Events

- New MISP Event
- Attribute has been saved
- New discussion post
- New user created
- Query against third-party services
- ...

❓ Supported events in MISP are called **Triggers**
❓ A **Trigger** is associated with **1-and-only-1 Workflow**



Triggers currently available

Currently 11 triggers can be hooked. 3 being Blocking.

Triggers

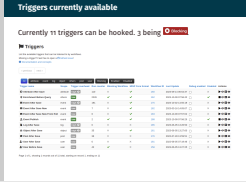
List the available triggers that can be listened to by workflows.
Missing a trigger? Feel free to open a Github issue!
Documentation and concepts

« previous next »

All | attribute | event | log | object | others | post | user | Blocking | Enabled | Disabled

Trigger name	Scope	Trigger overhead	Run counter	Blocking Workflow	MISP Core format	Workflow ID	Last Update	Debug enabled	Enabled	Actions
Attribute After Save	attribute	high	110	✗	✓	160	2023-09-14 06:54:37	☐	✗	
Enrichment Before Query	others	low	2226	✓	✓	162	2023-10-09 07:56:42	☐	✓	
Event After Save	event	high	191	✗	✓	175	2023-10-02 14:55:19	☐	✗	
Event After Save New	event	low	7	✗	✓	182	2023-03-16 14:05:07	☐	✗	
Event After Save New From Pull	event	low	6	✗	✓	183	2023-10-09 07:57:02	☐	✗	
Event Publish	event	low	2	✓	✓	188	2023-10-09 07:56:25	☐	✓	
Log After Save	log	high	0	✗	✗	185	2023-06-05 13:26:50	☐	✗	
Object After Save	object	high	35	✗	✓	161	2023-06-05 13:27:00	☐	✗	
Post After Save	post	low	36	✗	✗	176	2022-07-28 13:59:51	☐	✗	
User After Save	user	low	0	✗	✗	181	2022-08-05 07:19:46	☐	✗	
User Before Save	user	low	42	✓	✗	158	2023-06-05 13:27:25	☐	✗	

Page 1 of 1, showing 1 records out of 11 total, starting on record 1, ending on 11

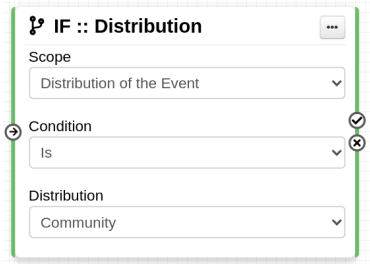


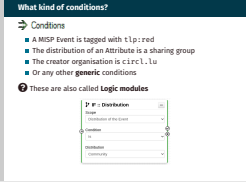
What kind of conditions?

Conditions

- A MISP Event is tagged with `tlp:red`
- The distribution of an Attribute is a sharing group
- The creator organisation is `circl.lu`
- Or any other **generic** conditions

These are also called **Logic modules**



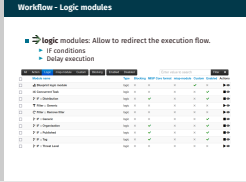


Workflow - Logic modules

- **logic** modules: Allow to redirect the execution flow.
  - IF conditions
  - Delay execution

All | Action | Logic | misp-module | Custom | Blocking | Enabled | Disabled — Enter value to search — Filter

Module name	Type	Blocking	MISP Core format	misp-module	Custom	Enabled	Actions
Blueprint logic module	logic	✗	✗	✗	✓	✗	▶ 👁
Concurrent Task	logic	✗	✗	✗	✗	✓	■ 👁
IF :: Distribution	logic	✗	✓	✗	✗	✓	■ 👁
Filter :: Generic	logic	✗	✗	✗	✗	✗	▶ 👁
Filter :: Remove filter	logic	✗	✗	✗	✗	✗	▶ 👁
IF :: Generic	logic	✗	✗	✗	✗	✓	■ 👁
IF :: Organisation	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Published	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Tag	logic	✗	✓	✗	✗	✓	■ 👁
IF :: Threat Level	logic	✗	✗	✗	✗	✗	▶ 👁

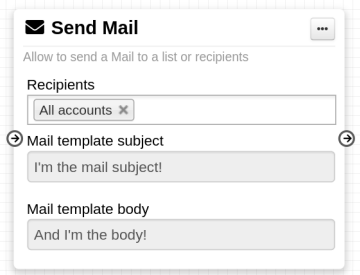


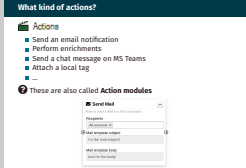
What kind of actions?

Actions

- Send an email notification
- Perform enrichments
- Send a chat message on MS Teams
- Attach a local tag
- ...

❓ These are also called **Action modules**



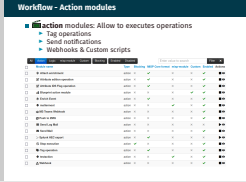


Workflow - Action modules

- **action** modules: Allow to executes operations
  - Tag operations
  - Send notifications
  - Webhooks & Custom scripts

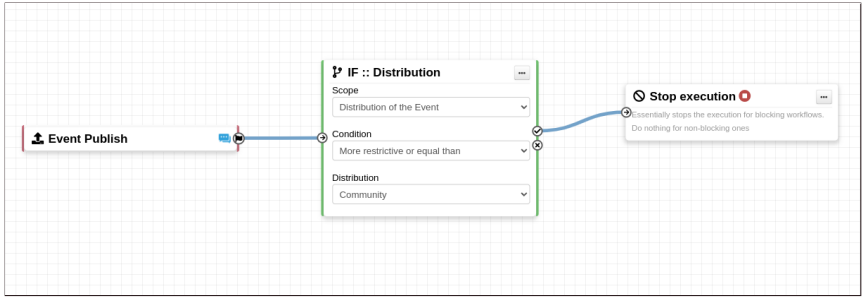
All | Action | Logic | misp-module | Custom | Blocking | Enabled | Disabled | Enter value to search | Filter

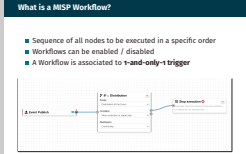
Module name	Type	Blocking	MISP Core format	misp-module	Custom	Enabled	Actions
Attach enrichment	action	✗	✓	✗	✗	✓	
Attribute edition operation	action	✗	✓	✗	✗	✓	
Attribute IDS Flag operation	action	✗	✓	✗	✗	✓	
Blueprint action module	action	✗	✗	✗	✓	✓	
Enrich Event	action	✗	✓	✗	✗	✓	
mattermost	action	✗	✗	✓	✗	✓	
MS Teams Webhook	action	✗	✗	✗	✗	✓	
Push to ZMQ	action	✗	✗	✗	✗	✓	
Send Log Mail	action	✗	✗	✗	✗	✗	
Send Mail	action	✗	✗	✗	✗	✓	
Splunk HEC export	action	✗	✓	✗	✗	✗	
Stop execution	action	✓	✗	✗	✗	✓	
Tag operation	action	✗	✓	✗	✗	✓	
testaction	action	✗	✗	✓	✗	✓	
Webhook	action	✗	✗	✗	✗	✓	



What is a MISP Workflow?

- Sequence of all nodes to be executed in a specific order
- Workflows can be enabled / disabled
- A Workflow is associated to **1-and-only-1 trigger**





Sources of Workflow modules (0)

Currently 36 built-in modules.

- **Trigger** module (11): built-in **only**
  - Get in touch if you want more
- **Logic** module (10): built-in & **custom**
- **Action** module (20): built-in & **custom**

Sources of Workflow modules (0)
Currently 36 built-in modules.
- **Trigger** module (11): built-in **only**
  - Get in touch if you want more
- **Logic** module (10): built-in & **custom**
- **Action** module (20): built-in & **custom**

Sources of Workflow modules (1)

- Built-in **default** modules
  - Part of the MISP codebase
  - Get in touch if you want us to increase the selection (or merge PR!)



Sources of Workflow modules (1)
Built-in default modules
Part of the MISP codebase
Get in touch if you want us to increase the selection (or merge PR!)

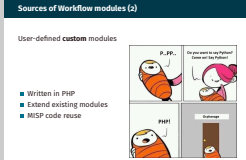


Sources of Workflow modules (2)

User-defined **custom** modules

- Written in PHP
- Extend existing modules
- MISP code reuse

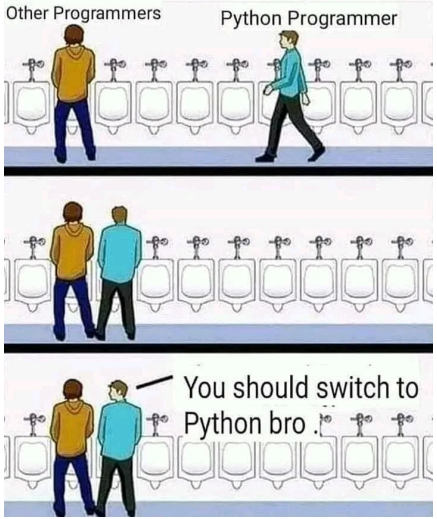


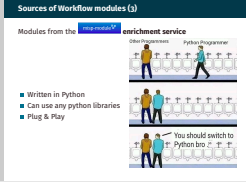


Sources of Workflow modules (3)

Modules from the misp-module **enrichment service**

- Written in Python
- Can use any python libraries
- Plug & Play





Demo by examples

WF-1. Send an email to **all admins** when a new event has been pulled

WF-2. Block queries on 3rd party services when **tlp:red** or **PAP:red**

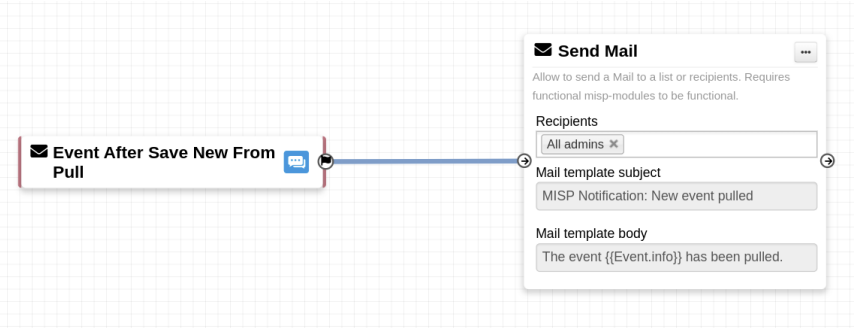
- **tlp:red**: For the eyes and ears of individual recipients only
- **PAP:RED**: Only passive actions that are not detectable from the outside

Demo by examples

WF-1. Send an email to **all admins** when a new event has been pulled

WF-2. Block queries on 3rd party services when **tlp:red** or **PAP:red**
- **tlp:red**: For the eyes and ears of individual recipients only
- **PAP:RED**: Only passive actions that are not detectable from the outside

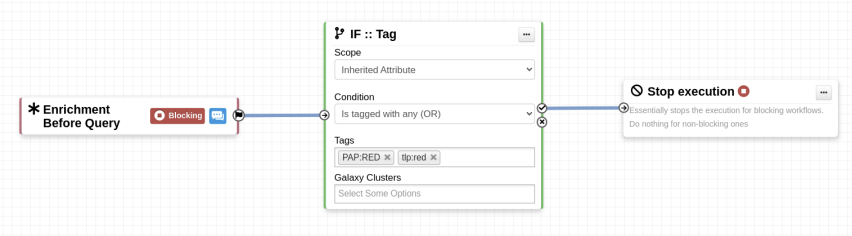
Demo WF-1: Send an email to all admins when a new event has been pulled

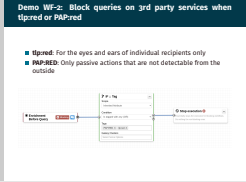




Demo WF-2: Block queries on 3rd party services when tlp:red or PAP:red

- **tlp:red**: For the eyes and ears of individual recipients only
- **PAP:RED**: Only passive actions that are not detectable from the outside





Everything is ready?

Let’s see how to build a workflow!



Getting started with workflows

Everything is ready?

Let’s see how to build a workflow!

Creating a workflow with the editor

1. Prevent event publication **if tlp:red** tag
   - Send a mail to `admin@admin.test` about potential data leak
2. **else**, send a notification on Mattermost

Creating a workflow with the editor
1. Prevent event publication **if tlp:red** tag
   - Send a mail to admin@admin.test about potential data leak
2. **else**, send a notification on Mattermost

Considerations when working with workflows

Objective: Overview of some common pitfalls



Considerations when

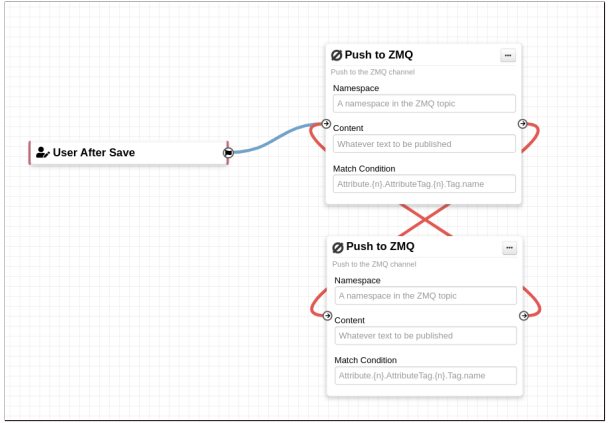
Considerations when working with workflows

Objective: Overview of some common pitfalls

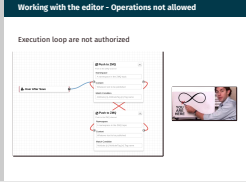


Working with the editor - Operations not allowed

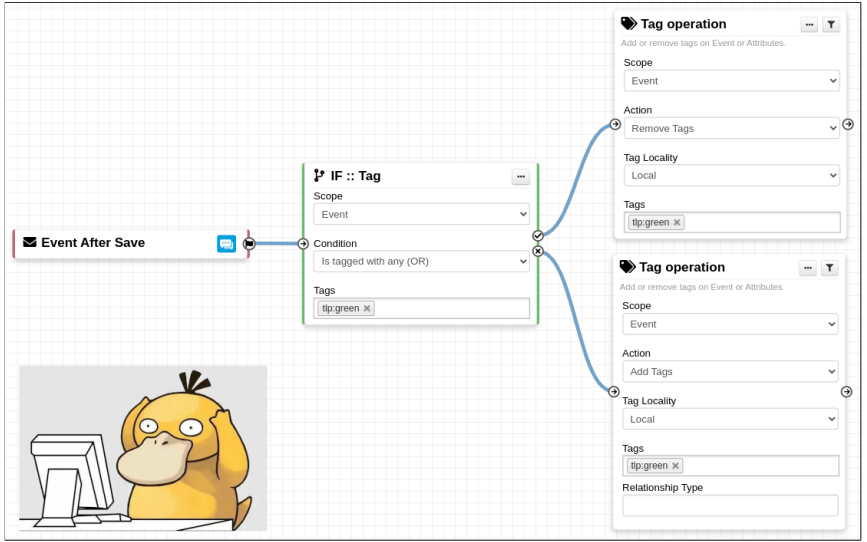
Execution loop are not authorized







Recursive workflows

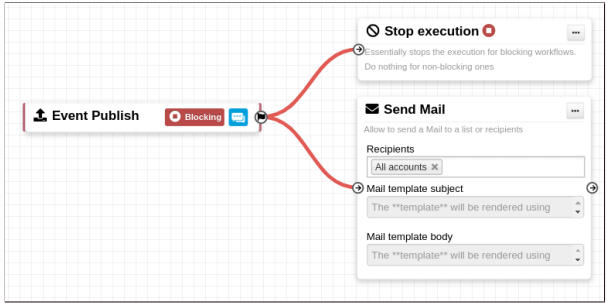


⚠Recursion: If an action re-run the workflow



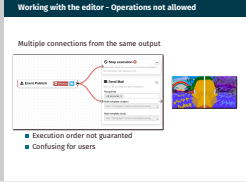
Working with the editor - Operations not allowed

Multiple connections from the same output





- Execution order not guaranted
- Confusing for users



Advanced usage

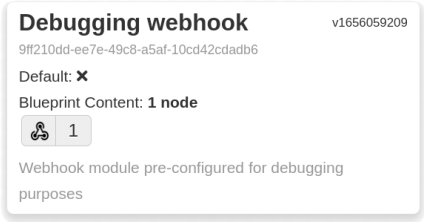
Objective: Overview of Blueprints, Data format and Filtering

Advanced usage

Objective: Overview of Blueprints, Data format and Filtering

Workflow blueprints

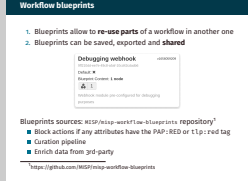
1. Blueprints allow to **re-use parts** of a workflow in another one
2. Blueprints can be saved, exported and **shared**



Blueprints sources: `MISP/misp-workflow-blueprints` repository[1]

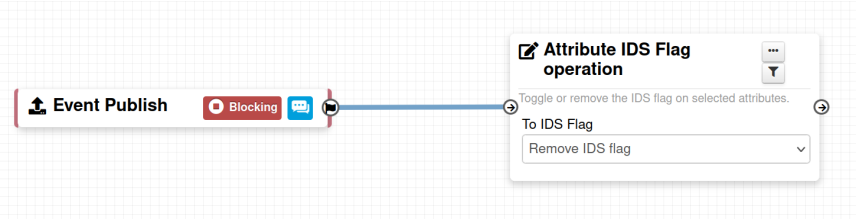
- Block actions if any attributes have the `PAP:RED` or `tlp:red` tag
- Curation pipeline
- Enrich data from 3rd-party

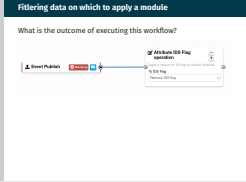
[1] https://github.com/MISP/misp-workflow-blueprints



Fitlering data on which to apply a module

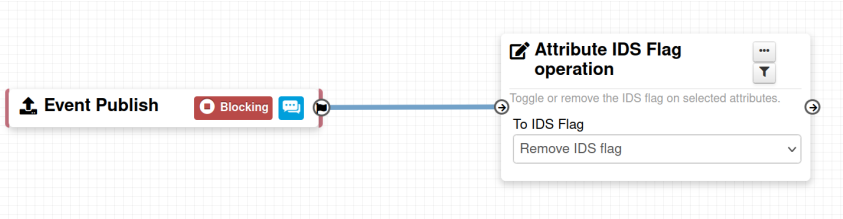
What is the outcome of executing this workflow?





Fitlering data on which to apply a module

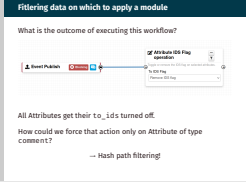
What is the outcome of executing this workflow?



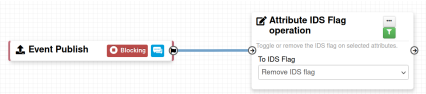
All Attributes get their `to_ids` turned off.

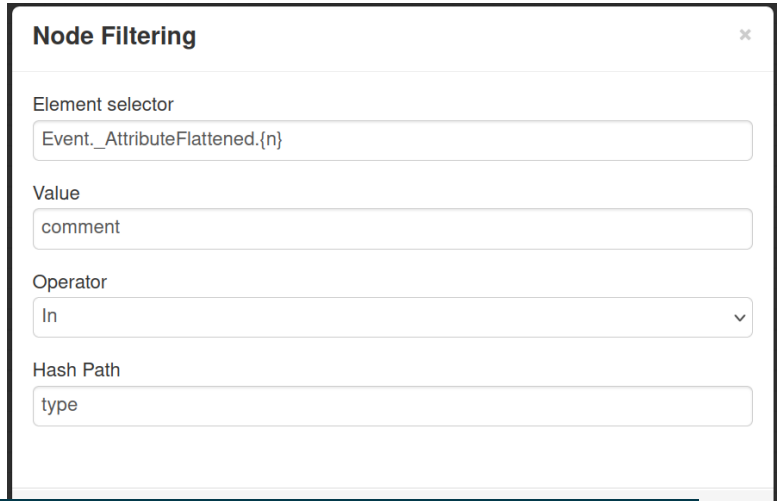
How could we force that action only on Attribute of type `comment`?

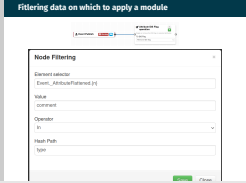
→ Hash path filtering!



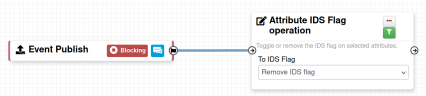
Fitlering data on which to apply a module

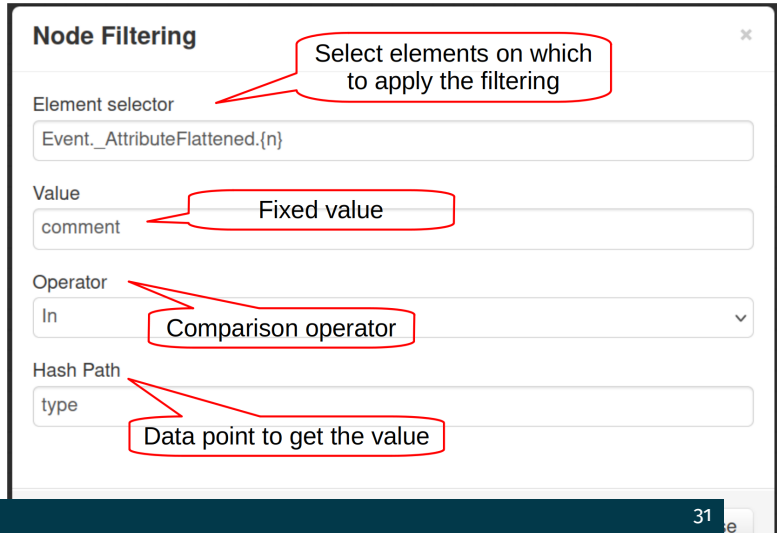


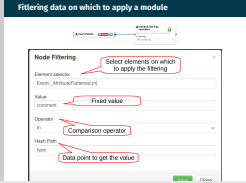




Fitlering data on which to apply a module

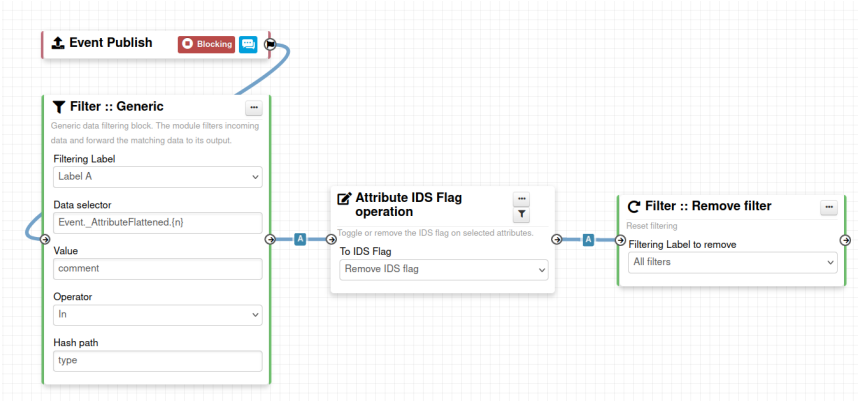


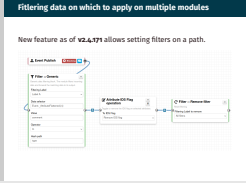




Fitlering data on which to apply on multiple modules

New feature as of **v2.4.171** allows setting filters on a path.





Should I migrate to MISP Workflows

I have automation in place using the API/ZMQ. Should I move to Workflows?

- I have a curation pipeline using the API, should I port it to workflows?
  - **No** in general, but WF can be used to start the curation process or perform simple pre-processing
- What if I want to **block** some actions
  - Put the blocking logic in the WF, keep the remaining outside
- Bottom line is **Keep it simple** for you to maintain

Should I migrate to MISP Workflows

I have automation in place using the API/ZMQ. Should I move to Workflows?

- I have a curation pipeline using the API, should I port it to workflows?
  - **No** in general, but WF can be used to start the curation process or perform simple pre-processing
- What if I want to **block** some actions
  - Put the blocking logic in the WF, keep the remaining outside
- Bottom line is **Keep it simple** for you to maintain

Future works

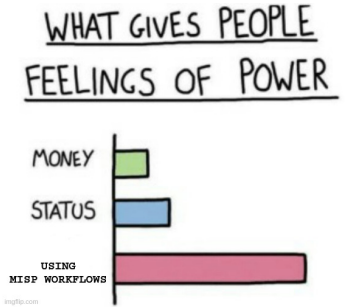
- More modules
- More modules
- More triggers
- Recursion prevention system



Future works
More modules
More modules
More triggers
Recursion prevention system
MISP WORKFLOWS
TODOS

Final words

- Designed to **quickly** and **cheaply** integrate MISP in CTI pipelines
- **Beta** Feature unlikely to change. But still..
- Waiting for feedback!
  - New triggers?
  - New modules?



Final words
Designed to quickly and cheaply integrate MISP in CTI pipelines
Beta Feature unlikely to change. But still..
Waiting for feedback!
New triggers?
New modules?
WHAT GIVES PEOPLE FEELINGS OF POWER
MONEY
STATUS